Lecture 2: Linear Algebra Review - Vector Norms

1. Vector Norms

We begin with the most basic way to measure the "size" of a vector.

The standard **Euclidean norm** (or **2-norm**) of a vector $\vec{x} \in \mathbb{R}^n$ is:

$$\|\vec{x}\| = \sqrt{\sum_{i=1}^{n} x_i^2}$$

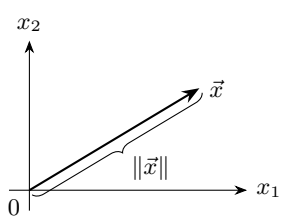
This is a special case of a more general concept.

1.1 Definition of a Norm

Definition 1 (Norm)**.** A function $\|\cdot\| : X \to \mathbb{R}$ is a **norm** if it satisfies three properties for all $\vec{x}, \vec{y} \in X$ and all $\alpha \in \mathbb{R}$:

1. **Non-negativity:** $\|\vec{x}\| \geq 0$, and $\|\vec{x}\| = 0$ if and only if $\vec{x} = \vec{0}$
2. **Triangle inequality:** $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$
3. **Absolute homogeneity:** $\|\alpha\vec{x}\| = |\alpha| \, \|\vec{x}\|$

These axioms capture our intuition about "length": it is always non-negative, the direct path is never longer than a detour, and scaling a vector scales its length proportionally.



Why each axiom matters. Positive definiteness (the "if and only if" part of non-negativity) prevents any nonzero vector from having zero size; without it, distinct vectors could be indistinguishable. **Homogeneity** says that scaling a vector by α scales its size by $|\alpha|$, so the norm is compatible with the linear structure of the space. **The triangle inequality** encodes the geometry we need for bounding errors, proving convergence of iterative algorithms, and establishing stability of solutions under perturbation.

Example 1 (Verifying the norm axioms for ℓ_2)**.** Let $\vec{x} = (1, -2)^\top$ and $\vec{y} = (2, 1)^\top$. We verify all three axioms for the ℓ_2 norm.

Positive definiteness. Since $\vec{x} \neq \vec{0}$, we need $\|\vec{x}\|_2 > 0$:

$$\|\vec{x}\|_2 = \sqrt{1^2 + (-2)^2} = \sqrt{5} > 0 \quad \checkmark$$

Homogeneity. Take $\alpha = -3$. We check $\|\alpha\vec{x}\|_2 = |\alpha| \, \|\vec{x}\|_2$:

$$\alpha\vec{x} = (-3, 6)^\top, \quad \|\alpha\vec{x}\|_2 = \sqrt{9 + 36} = 3\sqrt{5} = |-3| \cdot \sqrt{5} \quad \checkmark$$

Triangle inequality. We check $\|\vec{x} + \vec{y}\|_2 \leq \|\vec{x}\|_2 + \|\vec{y}\|_2$:

$$\vec{x} + \vec{y} = (3, -1)^\top, \quad \|\vec{x} + \vec{y}\|_2 = \sqrt{10}, \quad \|\vec{y}\|_2 = \sqrt{5}$$

$$\|\vec{x}+\vec{y}\|_2 = \sqrt{10} \approx 3.16 \leq 2\sqrt{5} \approx 4.47 = \|\vec{x}\|_2 + \|\vec{y}\|_2 \quad \checkmark$$

1.2 The ℓ_p Norm

The most important family of norms is the ℓ_p **norm**, parameterized by p.

Definition 2 (ℓ_p Norm)**.** For $1 \leq p < \infty$, the ℓ_p norm of $\vec{x} \in \mathbb{R}^n$ is:

$$\|\vec{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}$$

Different values of p give different norms, each emphasizing a different notion of size: $\|\vec{x}\|_1$ measures the total magnitude across all components, $\|\vec{x}\|_2$ measures the Euclidean length, and $\|\vec{x}\|_\infty$ measures the largest coordinate in absolute value.

$p = 2$**: Euclidean norm.** The familiar geometric distance:

$$\|\vec{x}\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2} = \sqrt{\vec{x}^\top \vec{x}}$$

$p = 1$**: Manhattan norm.** The sum of absolute values:

$$\|\vec{x}\|_1 = \sum_{i=1}^{n} |x_i|$$

This measures distance along axis-aligned paths (like walking on a grid of city blocks).

Example 2 (Comparing norms on a single vector)**.** Let $\vec{x} = (3, -4)^\top$. The three standard ℓ_p norms give different values:

$$\|\vec{x}\|_1 = |3|+|-4| = 7, \qquad \|\vec{x}\|_2 = \sqrt{9+16} = 5, \qquad \|\vec{x}\|_\infty = 4$$

$$\|\vec{x}\|_\infty = 4 \;\le\; \|\vec{x}\|_2 = 5 \;\le\; \|\vec{x}\|_1 = 7$$

The ordering $\|\vec{x}\|_\infty \le \|\vec{x}\|_2 \le \|\vec{x}\|_1$ holds for every $\vec{x} \in \mathbb{R}^n$.

$p = \infty$**: Max norm.** The largest absolute component:

$$\|\vec{x}\|_\infty = \max_{i=1,2,\ldots,n} |x_i|$$

This is the limiting case as $p \to \infty$. Only the largest component matters.

1.3 The ℓ^∞ Norm as a Limit

The ℓ_∞ norm is not just a separate definition—it arises naturally as the limit of ℓ_p norms.

Claim 1. *For any $\vec{x} \in \mathbb{R}^n$, $\|\vec{x}\|_\infty = \lim_{p\to\infty} \|\vec{x}\|_p$.*

Proof. Let $M = \max_i |x_i| = \|\vec{x}\|_\infty$. We use a squeeze argument.

Lower bound. The sum includes a term equal to M^p, so $\|\vec{x}\|_p \ge (M^p)^{1/p} = M$.

Upper bound. Since $|x_i| \le M$ for every i, we have $\|\vec{x}\|_p \le (nM^p)^{1/p} = n^{1/p}M$.

Squeeze. Combining:

$$M \;\le\; \|\vec{x}\|_p \;\le\; n^{1/p} \cdot M$$

As $p \to \infty$, $n^{1/p} \to 1$, so both sides converge to $M = \|\vec{x}\|_\infty$. □

Example 3 (Convergence of $\|\vec{x}\|_p$ to $\|\vec{x}\|_\infty$)**.** Let $\vec{x} = (1, 2)^\top$, so $\|\vec{x}\|_\infty = 2$.

p	$\|\vec{x}\|_p$	Value
1	$\|1\| + \|2\|$	3.000
2	$\sqrt{1+4}$	2.236
4	$(1+16)^{1/4}$	2.031
10	$(1+1024)^{1/10}$	2.003
∞	$\max(1, 2)$	2.000

As p increases, $\|\vec{x}\|_p$ decreases monotonically toward $\|\vec{x}\|_\infty = 2$.

Rigor note: For $0 < p < 1$, the formula $\left(\sum_i |x_i|^p\right)^{1/p}$ does *not* define a norm because the triangle inequality fails. For example, with $p = 1/2$ and $\vec{e}_1, \vec{e}_2 \in \mathbb{R}^2$:

$$\|\vec{e}_1 + \vec{e}_2\|_{1/2} = (1+1)^2 = 4 > 2 = \|\vec{e}_1\|_{1/2} + \|\vec{e}_2\|_{1/2}$$

This is why we require $p \ge 1$.

2. Inner Product Inequalities

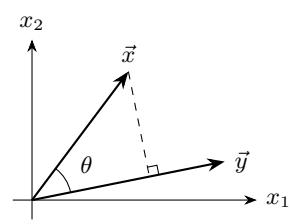
2.1 Cauchy–Schwarz Inequality

The most fundamental inequality relating inner products and norms:

Theorem 1 (Cauchy–Schwarz Inequality)**.** *For all $\vec{x}, \vec{y} \in \mathbb{R}^n$:*

$$|\vec{x}^\top \vec{y}| \le \|\vec{x}\|_2 \, \|\vec{y}\|_2$$

with equality if and only if $\vec{x}$ and $\vec{y}$ are linearly dependent.



Proof. If $\vec{y} = \vec{0}$, both sides are zero and the inequality holds trivially. So assume $\vec{y} \ne \vec{0}$. For any $t \in \mathbb{R}$, the squared norm is non-negative:

$$0 \le \|\vec{x} - t\vec{y}\|_2^2 = (\vec{x} - t\vec{y})^\top (\vec{x} - t\vec{y}) = \|\vec{x}\|_2^2 - 2t(\vec{x}^\top \vec{y}) + t^2 \|\vec{y}\|_2^2$$

The right-hand side is a quadratic in t that is non-negative for all $t \in \mathbb{R}$. A quadratic $At^2 + Bt + C \ge 0$ for all t if and only if its discriminant satisfies $B^2 - 4AC \le 0$. Here $A = \|\vec{y}\|_2^2$, $B = -2(\vec{x}^\top \vec{y})$, and $C = \|\vec{x}\|_2^2$, so:

$$\left(-2(\vec{x}^\top \vec{y})\right)^2 - 4\|\vec{y}\|_2^2 \|\vec{x}\|_2^2 \le 0$$

$$4(\vec{x}^\top \vec{y})^2 \le 4\|\vec{x}\|_2^2 \|\vec{y}\|_2^2$$

Dividing by 4 gives $(\vec{x}^\top \vec{y})^2 \le \|\vec{x}\|_2^2 \, \|\vec{y}\|_2^2$. Taking square roots:

$$|\vec{x}^\top \vec{y}| \leq \|\vec{x}\|_2 \, \|\vec{y}\|_2$$

□

Equality condition. Equality holds if and only if $\vec{x}$ and $\vec{y}$ are linearly dependent, i.e., one is a scalar multiple of the other. This is because the discriminant equals zero precisely when $\|\vec{x} - t\vec{y}\|_2^2 = 0$ for some t, meaning $\vec{x} = t\vec{y}$.

Example 4 (Cauchy–Schwarz with specific vectors)**.** Let $\vec{x} = (1, 2)^\top$ and $\vec{y} = (3, -1)^\top$. We compute:

$$\vec{x}^\top \vec{y} = (1)(3) + (2)(-1) = 1$$

$$\|\vec{x}\|_2 = \sqrt{1^2 + 2^2} = \sqrt{5}, \qquad \|\vec{y}\|_2 = \sqrt{3^2 + (-1)^2} = \sqrt{10}$$

The inequality gives:

$$|\vec{x}^\top \vec{y}| = 1 \leq \sqrt{5} \cdot \sqrt{10} = \sqrt{50} \approx 7.07$$

Equality does *not* hold here, since $\vec{x}$ and $\vec{y}$ are not scalar multiples of each other (there is no α with $(1, 2)^\top = \alpha(3, -1)^\top$).

Cauchy–Schwarz says that the inner product of two vectors is bounded by the product of their lengths. This is the algebraic version of the geometric fact that $|\cos\theta| \leq 1$.

2.2 Hölder's Inequality

Cauchy–Schwarz is a special case of a more general result.

Theorem 2 (Hölder's Inequality)**.** *Let $\vec{x}, \vec{y} \in \mathbb{R}^n$ and let $p, q \geq 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$. Then:*

$$|\vec{x}^\top \vec{y}| \leq \|\vec{x}\|_p \, \|\vec{y}\|_q$$

The pair (p, q) satisfying $\frac{1}{p} + \frac{1}{q} = 1$ are called **conjugate exponents**. Important cases:

- $p = q = 2$: recovers Cauchy–Schwarz
- $p = 1, q = \infty$: gives $|\vec{x}^\top \vec{y}| \leq \|\vec{x}\|_1 \, \|\vec{y}\|_\infty$
- $p = \infty, q = 1$: gives $|\vec{x}^\top \vec{y}| \leq \|\vec{x}\|_\infty \, \|\vec{y}\|_1$

Example 5 (Hölder with $p = 1$, $q = \infty$)**.** Let $\vec{x} = (1, -1, 0)^\top$ and $\vec{y} = (4, -4, 2)^\top$. We compute:

$$\sum_{i=1}^{3} |x_i y_i| = |1 \cdot 4| + |(-1)(-4)| + |0 \cdot 2| = 4 + 4 + 0 = 8$$

$$\|\vec{x}\|_1 = |1| + |-1| + |0| = 2, \qquad \|\vec{y}\|_\infty = \max(|4|, |-4|, |2|) = 4$$

So $|\vec{x}^\top \vec{y}| = 8 = 2 \cdot 4 = \|\vec{x}\|_1 \, \|\vec{y}\|_\infty$. This is an equality case: the bound is tight because each nonzero x_i is paired with a y_i achieving the maximum absolute value.

Proof of Hölder's Inequality

We prove the three cases separately.

Endpoint case $p = 1$, $q = \infty$.

Proof. By the triangle inequality for sums and the definition of the ℓ_∞ norm:

$$|\vec{x}^\top \vec{y}| = \left| \sum_{i=1}^{n} x_i y_i \right| \leq \sum_{i=1}^{n} |x_i y_i| = \sum_{i=1}^{n} |x_i||y_i|$$

Since $|y_i| \leq \max_j |y_j| = \|\vec{y}\|_\infty$ for every i, we can bound:

$$\sum_{i=1}^{n} |x_i||y_i| \leq \|\vec{y}\|_\infty \sum_{i=1}^{n} |x_i| = \|\vec{x}\|_1 \, \|\vec{y}\|_\infty$$

Combining the two chains:

$$|\vec{x}^\top \vec{y}| \leq \sum_{i=1}^{n} |x_i||y_i| \leq \left(\max_i |y_i|\right) \sum_{i=1}^{n} |x_i| = \|\vec{x}\|_1 \, \|\vec{y}\|_\infty$$

□

Endpoint case $p = \infty$, $q = 1$.

Proof. The argument is symmetric. We have:

$$|\vec{x}^\top \vec{y}| \leq \sum_{i=1}^{n} |x_i||y_i| \leq \left(\max_i |x_i|\right) \sum_{i=1}^{n} |y_i| = \|\vec{x}\|_\infty \, \|\vec{y}\|_1$$

□

General case $1 < p < \infty$.

The proof relies on a classical pointwise inequality.

Lemma 3 (Young's Inequality)**.** *For $a, b \geq 0$ and conjugate exponents $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$:*

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

Young's inequality follows from the concavity of the logarithm: $\log(ab) = \frac{1}{p}\log(a^p) + \frac{1}{q}\log(b^q) \leq \log\left(\frac{a^p}{p} + \frac{b^q}{q}\right)$.

Proof of Hölder's inequality for $1 < p < \infty$. If $\|\vec{x}\|_p = 0$ or $\|\vec{y}\|_q = 0$, the inequality holds trivially (both sides are zero). So assume both norms are positive.

Step 1: Normalize. Define:

$$u_i = \frac{|x_i|}{\|\vec{x}\|_p}, \qquad v_i = \frac{|y_i|}{\|\vec{y}\|_q}$$

By construction, the normalized vectors satisfy:

$$\sum_{i=1}^n u_i^p = \frac{1}{\|\vec{x}\|_p^p} \sum_{i=1}^n |x_i|^p = 1, \qquad \sum_{i=1}^n v_i^q = \frac{1}{\|\vec{y}\|_q^q} \sum_{i=1}^n |y_i|^q = 1$$

Step 2: Apply Young's inequality. For each $i = 1, \ldots, n$, apply Young's inequality to $a = u_i$ and $b = v_i$:

$$u_i \, v_i \leq \frac{u_i^p}{p} + \frac{v_i^q}{q}$$

Step 3: Sum over all components.

$$\sum_{i=1}^n u_i \, v_i \leq \frac{1}{p} \sum_{i=1}^n u_i^p + \frac{1}{q} \sum_{i=1}^n v_i^q = \frac{1}{p} \cdot 1 + \frac{1}{q} \cdot 1 = \frac{1}{p} + \frac{1}{q} = 1$$

Step 4: Undo the normalization. Substituting back:

$$\sum_{i=1}^n \frac{|x_i|}{\|\vec{x}\|_p} \cdot \frac{|y_i|}{\|\vec{y}\|_q} \leq 1$$

Multiplying both sides by $\|\vec{x}\|_p \, \|\vec{y}\|_q$:

$$\sum_{i=1}^n |x_i||y_i| \leq \|\vec{x}\|_p \, \|\vec{y}\|_q$$

Finally, since $|\vec{x}^\top \vec{y}| \leq \sum_{i=1}^n |x_i||y_i|$, we conclude:

$$|\vec{x}^\top \vec{y}| \leq \|\vec{x}\|_p \, \|\vec{y}\|_q$$

□

2.3 Minkowski's Inequality

Theorem 4 (Minkowski's Inequality)**.** *For* $1 \leq p \leq \infty$ *and* $\vec{u}, \vec{v} \in \mathbb{R}^n$*:*

$$\|\vec{u} + \vec{v}\|_p \leq \|\vec{u}\|_p + \|\vec{v}\|_p$$

This is the triangle inequality for ℓ_p norms. For $p = 1$ and $p = \infty$ it is straightforward. For $1 < p < \infty$ it follows from Hölder's inequality.

Proof sketch for $1 < p < \infty$. We expand $\|\vec{u} + \vec{v}\|_p^p$ and apply the triangle inequality componentwise:

$$\|\vec{u} + \vec{v}\|_p^p = \sum_{i=1}^n |u_i + v_i|^p = \sum_{i=1}^n |u_i + v_i|^{p-1} |u_i + v_i|$$

$$\leq \sum_{i=1}^n |u_i| \, |u_i + v_i|^{p-1} + \sum_{i=1}^n |v_i| \, |u_i + v_i|^{p-1}$$

Apply Hölder's inequality to each sum with exponents p and $q = \frac{p}{p-1}$. Note that the vector with components $|u_i + v_i|^{p-1}$ has ℓ_q norm:

$$\left(\sum_{i=1}^n \left(|u_i + v_i|^{p-1} \right)^q \right)^{1/q} = \left(\sum_{i=1}^n |u_i + v_i|^p \right)^{1/q} = \|\vec{u} + \vec{v}\|_p^{p/q}$$

since $(p-1)q = p$. Applying Hölder to both sums:

$$\|\vec{u} + \vec{v}\|_p^p \leq \|\vec{u}\|_p \, \|\vec{u} + \vec{v}\|_p^{p/q} + \|\vec{v}\|_p \, \|\vec{u} + \vec{v}\|_p^{p/q}$$

$$= \left(\|\vec{u}\|_p + \|\vec{v}\|_p \right) \|\vec{u} + \vec{v}\|_p^{p/q}$$

If $\|\vec{u} + \vec{v}\|_p = 0$, the inequality holds trivially. Otherwise, divide both sides by $\|\vec{u} + \vec{v}\|_p^{p/q}$. Since $p - p/q = p(1 - 1/q) = p \cdot \frac{1}{p} = 1$:

$$\|\vec{u} + \vec{v}\|_p \leq \|\vec{u}\|_p + \|\vec{v}\|_p$$

□

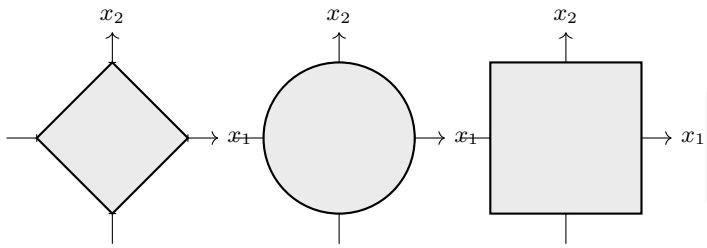
Remark. Minkowski's inequality proves that the ℓ_p formula from Definition 2 really does satisfy the norm axioms for all $p \geq 1$. Non-negativity and absolute homogeneity are immediate from the definition; the triangle inequality is the nontrivial property, and it is precisely Minkowski's inequality.

3. Norm Balls

Definition 3 (Norm Ball)**.** The **unit norm ball** in the ℓ_p norm is the set of all vectors with norm at most 1:

$$B_p = \{\vec{x} \in \mathbb{R}^n : \|\vec{x}\|_p \leq 1\}$$

The shape of B_p depends on p. In $\mathbb{R}^2$:



B_1: $p = 1$ B_2: $p = 2$ B_∞: $p = \infty$

- $p = 2$: circle (all points at Euclidean distance ≤ 1)
- $p = 1$: diamond (constraint $|x_1| + |x_2| \leq 1$)
- $p = \infty$: square (constraint $\max(|x_1|, |x_2|) \leq 1$)

Key observation: As p increases from 1 to ∞, the norm ball "inflates" from a diamond through a circle to a square. Smaller p produces "spikier" balls.

4. Optimization Over Norm Balls

A natural question: given a fixed vector $\vec{y}$, what is the maximum inner product $\vec{x}^\top \vec{y}$ over all $\vec{x}$ in the unit ℓ_p ball?

$$\max_{\|\vec{x}\|_p \leq 1} \vec{x}^\top \vec{y}$$

The answer depends on p and reveals a deep connection between norms and their conjugates.

4.1 Upper Bound via Hölder

Before examining individual cases, we establish a universal upper bound.

For any $\vec{x}$ with $\|\vec{x}\|_p \leq 1$, Hölder's inequality gives:

$$\vec{x}^\top \vec{y} \leq |\vec{x}^\top \vec{y}| \leq \|\vec{x}\|_p \, \|\vec{y}\|_q \leq 1 \cdot \|\vec{y}\|_q = \|\vec{y}\|_q$$

where q is the conjugate exponent satisfying $\frac{1}{p} + \frac{1}{q} = 1$.

Upper bound: $\max_{\|\vec{x}\|_p \leq 1} \vec{x}^\top \vec{y} \leq \|\vec{y}\|_q$ for *every* $p \geq 1$.

So the maximum is *at most* $\|\vec{y}\|_q$. To show equality, it suffices to exhibit, for each p, a feasible $\vec{x}^\star$ with $\|\vec{x}^\star\|_p \leq 1$ achieving $\vec{x}^{\star\top} \vec{y} = \|\vec{y}\|_q$.

4.2 Case $p = 2$: Euclidean Ball

Example 6 ($p = 2$)**.** Maximize $\vec{x}^\top \vec{y}$ subject to $\|\vec{x}\|_2 \leq 1$, with $\vec{y} = (1, 2)^\top$.

The inner product $\vec{x}^\top \vec{y}$ is maximized when $\vec{x}$ points in the same direction as $\vec{y}$. The optimal solution is:

$$\vec{x}^\star = \frac{\vec{y}}{\|\vec{y}\|_2}$$

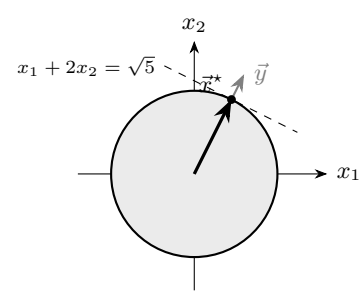
Verification. We have $\|\vec{x}^\star\|_2 = 1$ (feasible), and:

$$\vec{x}^{\star\top} \vec{y} = \frac{\vec{y}^\top \vec{y}}{\|\vec{y}\|_2} = \frac{\|\vec{y}\|_2^2}{\|\vec{y}\|_2} = \|\vec{y}\|_2$$

This matches the upper bound ($q = 2$ when $p = 2$), so $\vec{x}^\star$ is optimal.

Result: $\max_{\|\vec{x}\|_2 \leq 1} \vec{x}^\top \vec{y} = \|\vec{y}\|_2$. This is precisely Cauchy–Schwarz with equality.

Geometrically: on the circle, the best point is the point in the direction of $\vec{y}$.



Example 7 (Numerical: $p = 2$)**.** Let $\vec{y} = (3, 4)^\top$. Then $\|\vec{y}\|_2 = \sqrt{9 + 16} = 5$. The maximizer is:

$$\vec{x}^\star = \frac{1}{5}(3, 4)^\top = \left(\tfrac{3}{5}, \tfrac{4}{5}\right)^\top$$

Verify: $\|\vec{x}^\star\|_2 = \sqrt{9/25 + 16/25} = 1$, and $\vec{x}^{\star\top} \vec{y} = \frac{9}{5} + \frac{16}{5} = 5 = \|\vec{y}\|_2$. ✓

4.3 Case $p = 1$: Diamond Ball

Example 8 ($p = 1$)**.** Maximize $\vec{x}^\top \vec{y}$ subject to $\|\vec{x}\|_1 \leq 1$.

The constraint is $|x_1| + |x_2| + \cdots + |x_n| \leq 1$. The inner product is:

$$\vec{x}^\top \vec{y} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

To maximize, we should put all our "budget" of $\|\vec{x}\|_1 \leq 1$ into the component where $|y_i|$ is largest. Let $i^\star = \operatorname{argmax}_i |y_i|$. Then choose:

$$x_{i^\star} = \operatorname{sign}(y_{i^\star}), \quad x_j = 0 \text{ for } j \neq i^\star$$

That is, $\vec{x}^\star = \operatorname{sign}(y_{i^\star}) \, \vec{e}_{i^\star}$.

Verification. We have $\|\vec{x}^\star\|_1 = 1$ (feasible), and:

$$\vec{x}^{\star\top}\vec{y} = |y_{i^\star}| = \max_i |y_i| = \|\vec{y}\|_\infty$$

Formal proof. For any $\vec{x}$ with $\|\vec{x}\|_1 \leq 1$:

$$\vec{x}^\top\vec{y} = \sum_{i=1}^n x_i y_i \leq \sum_{i=1}^n |x_i||y_i| \leq \sum_{i=1}^n |x_i| \cdot |y_{\max}|$$

$$= |y_{\max}| \sum_{i=1}^n |x_i| \leq |y_{\max}| \cdot 1 = \|\vec{y}\|_\infty$$

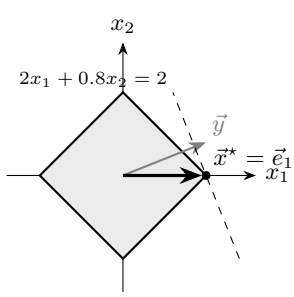
Example 9 (Numerical: $p = 1$)**.** Let $\vec{y} = (2, -5, 1)^\top$. The largest absolute component is $|y_2| = 5$, so $i^\star = 2$:

$$\vec{x}^\star = \text{sign}(-5)\,\vec{e}_2 = (0, -1, 0)^\top$$

Verify: $\|\vec{x}^\star\|_1 = 1$, and $\vec{x}^{\star\top}\vec{y} = 0 + 5 + 0 = 5 = \|\vec{y}\|_\infty$. ✓

> **Result:** $\max_{\|\vec{x}\|_1 \leq 1} \vec{x}^\top\vec{y} = \|\vec{y}\|_\infty$. The conjugate of the ℓ_1 norm is the ℓ_∞ norm.

Geometric meaning: The ℓ_1-ball is a diamond. A linear objective is maximized at a corner of the diamond.



4.4 Case $p = \infty$: Cube Ball

Example 10 ($p = \infty$)**.** Maximize $\vec{x}^\top\vec{y}$ subject to $\|\vec{x}\|_\infty \leq 1$, with $\vec{y} = (-2, 3)^\top$.

The constraint $\|\vec{x}\|_\infty \leq 1$ means each component satisfies $x_i \in [-1, 1]$. To maximize $\vec{x}^\top\vec{y} = \sum_i x_i y_i$, choose each x_i independently to maximize $x_i y_i$:

$$x_i^\star = \text{sign}(y_i)$$

For $\vec{y} = (-2, 3)^\top$: $\vec{x}^\star = (-1, 1)^\top$, giving $\vec{x}^{\star\top}\vec{y} = (-1)(-2) + (1)(3) = 5 = |-2| + |3| = \|\vec{y}\|_1$.

Formal proof. For any $\vec{x}$ with $\|\vec{x}\|_\infty \leq 1$:

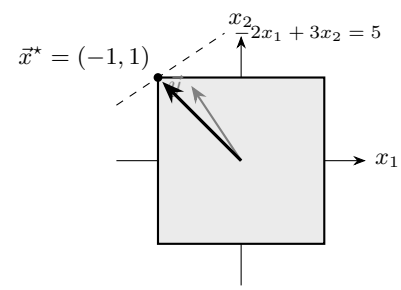
$$\vec{x}^\top\vec{y} = \sum_{i=1}^n x_i y_i \leq \sum_{i=1}^n |x_i||y_i| \leq \sum_{i=1}^n |y_i| = \|\vec{y}\|_1$$

Example 11 (Numerical: $p = \infty$)**.** Let $\vec{y} = (2, -5, 1)^\top$. Then $\vec{x}^\star = (1, -1, 1)^\top$, $\|\vec{x}^\star\|_\infty = 1$, and:

$$\vec{x}^{\star\top}\vec{y} = 2 + 5 + 1 = 8 = |2| + |-5| + |1| = \|\vec{y}\|_1 ✓$$

> **Result:** $\max_{\|\vec{x}\|_\infty \leq 1} \vec{x}^\top\vec{y} = \|\vec{y}\|_1$. The conjugate of the ℓ_∞ norm is the ℓ_1 norm.

Geometric meaning: The ℓ_∞-ball is a square. The best point is the corner whose signs match the signs of $\vec{y}$.



4.5 General Maximizer for $1 < p < \infty$

For general $p \in (1, \infty)$ with conjugate exponent q satisfying $\frac{1}{p} + \frac{1}{q} = 1$, the optimal solution is:

$$x_i^\star = \frac{\text{sign}(y_i)\,|y_i|^{q-1}}{\|\vec{y}\|_q^{q-1}}$$

Verification of feasibility ($\|\vec{x}^\star\|_p = 1$). Using the identity $p(q-1) = q$:

$$\|\vec{x}^\star\|_p^p = \sum_{i=1}^n |x_i^\star|^p = \sum_{i=1}^n \frac{|y_i|^{p(q-1)}}{\|\vec{y}\|_q^{p(q-1)}} = \frac{1}{\|\vec{y}\|_q^q} \sum_{i=1}^n |y_i|^q = \frac{\|\vec{y}\|_q^q}{\|\vec{y}\|_q^q} = 1$$

Verification of optimality ($\vec{x}^{\star\top}\vec{y} = \|\vec{y}\|_q$):

$$\vec{x}^{\star\top}\vec{y} = \sum_{i=1}^n \frac{\text{sign}(y_i)\,|y_i|^{q-1}}{\|\vec{y}\|_q^{q-1}} \cdot y_i = \frac{1}{\|\vec{y}\|_q^{q-1}} \sum_{i=1}^n |y_i|^q = \frac{\|\vec{y}\|_q^q}{\|\vec{y}\|_q^{q-1}} = \|\vec{y}\|_q$$

This matches the Hölder upper bound, confirming optimality for all $1 < p < \infty$.

> **Key identity used:** Since $\frac{1}{p} + \frac{1}{q} = 1$, we have $q = \frac{p}{p-1}$, so $q - 1 = \frac{1}{p-1}$ and $p(q-1) = \frac{p}{p-1} = q$.

Example 12 (Numerical: $p = 3$)**.** Let $p = 3$, so $q = \frac{3}{2}$. Let $\vec{y} = (1, 1)^\top$. Then $\|\vec{y}\|_{3/2} = (1 + 1)^{2/3} = 2^{2/3}$.

Using $q - 1 = \frac{1}{2}$, the maximizer is $x_i^\star = 1^{1/2}/(2^{2/3})^{1/2} = 2^{-1/3}$, so $\vec{x}^\star = 2^{-1/3}(1, 1)^\top$.

Feasibility: $\|\vec{x}^\star\|_3 = (2^{-1} + 2^{-1})^{1/3} = 1^{1/3} = 1$. ✓

Optimality: $\vec{x}^{\star\top}\vec{y} = 2 \cdot 2^{-1/3} = 2^{2/3} = \|\vec{y}\|_{3/2}$. ✓

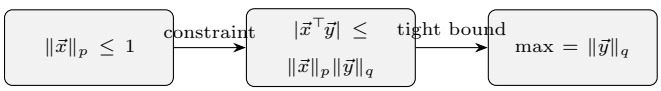
4.6 The Dual Norm Pattern

The cases above reveal a beautiful pattern. In each case, optimizing over the ℓ_p ball yields the conjugate ℓ_q norm:

$$\max_{\|\vec{x}\|_p \leq 1} \vec{x}^\top \vec{y} = \|\vec{y}\|_q \quad \text{where } \frac{1}{p} + \frac{1}{q} = 1$$

The norm $\|\cdot\|_q$ is called the **dual norm** of $\|\cdot\|_p$. Hölder's inequality provides the upper bound, and the explicit constructions above show the bound is tight for every $p \geq 1$.

p	q (conjugate)	Dual norm
1	∞	$\|\cdot\|_\infty$
2	2	$\|\cdot\|_2$
∞	1	$\|\cdot\|_1$



$$p = 1 \leftrightarrow q = \infty \qquad p = 2 \leftrightarrow q = 2 \qquad p = \infty \leftrightarrow q = 1$$

The ℓ_2 norm is self-dual ($q = 2$ when $p = 2$), which is one reason it is so special.

4.7 Problem Solving Strategy: Separability

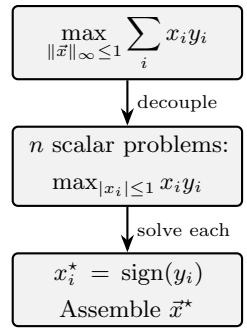
Strategy (Separability). When trying to simplify an optimization problem, check whether it decomposes into independent scalar problems. Solve each separately, then assemble the vector solution.

This is the structure of the $p = \infty$ case. The constraint $\|\vec{x}\|_\infty \leq 1$ means $-1 \leq x_i \leq 1$ for each component *independently*, so the vector problem

$$\max_{\|\vec{x}\|_\infty \leq 1} \sum_{i=1}^{n} x_i y_i$$

decomposes into n independent scalar problems $\max_{-1 \leq x_i \leq 1} x_i y_i$, each solved by $x_i^\star = \text{sign}(y_i)$.

When to use this: Separability applies when the constraint set is a Cartesian product of per-component constraints *and* the objective is a sum of per-component terms. The ℓ_∞ ball is $[-1, 1]^n$, so this works. The ℓ_1 and ℓ_2 balls couple the components, so separability does not apply.



4.8 Problem Solving Strategy: Bound and Attain

Strategy (Bound and Attain). To solve an optimization problem:

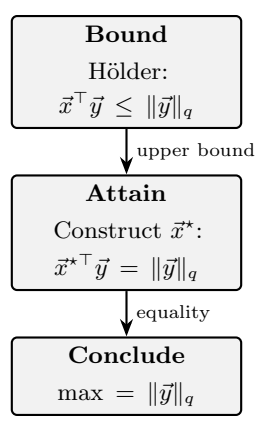
1. **Bound:** Use an inequality to establish an upper bound on the objective over all feasible points.
2. **Attain:** Construct a specific feasible point that achieves the upper bound with equality.

This is the proof structure we used for the $p = 1$ case and the general dual norm result. First, Hölder gives the bound:

$$\vec{x}^\top \vec{y} \leq \|\vec{x}\|_p \, \|\vec{y}\|_q \leq \|\vec{y}\|_q$$

Then we constructed an explicit $\vec{x}^\star$ achieving equality. Together, the two steps prove that the optimum is exactly $\|\vec{y}\|_q$.

Why this matters: "Bound and attain" is one of the core proof templates in optimization. Many results in this course—including duality theory, minimax theorems, and optimality conditions—follow this two-step pattern.



5. Gram–Schmidt and QR Decomposition

Given a set of linearly independent vectors $\{\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_m\}$, the **Gram–Schmidt process** produces an orthonormal set $\{\vec{q}_1, \vec{q}_2, \ldots, \vec{q}_m\}$ spanning the same subspace.

5.1 The Gram–Schmidt Process

Step 1: Normalize the first vector.

$$\vec{q}_1 = \frac{\vec{a}_1}{\|\vec{a}_1\|_2}$$

Step 2: Orthogonalize and normalize $\vec{a}_2$.

Project $\vec{a}_2$ onto $\vec{q}_1$ and subtract:

$$\vec{s}_2 = \vec{a}_2 - \langle \vec{a}_2, \vec{q}_1 \rangle \vec{q}_1$$

The residual $\vec{s}_2$ is orthogonal to $\vec{q}_1$. Normalize:

$$\vec{q}_2 = \frac{\vec{s}_2}{\|\vec{s}_2\|_2}$$

Step 3: Orthogonalize and normalize $\vec{a}_3$.

Project $\vec{a}_3$ onto $\vec{q}_1$ and $\vec{q}_2$, then subtract both:

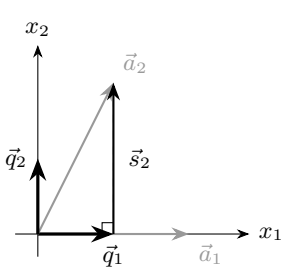
$$\vec{s}_3 = \vec{a}_3 - \langle \vec{a}_3, \vec{q}_1 \rangle \vec{q}_1 - \langle \vec{a}_3, \vec{q}_2 \rangle \vec{q}_2$$

Normalize: $\vec{q}_3 = \vec{s}_3 / \|\vec{s}_3\|_2$.

General step k: Subtract projections onto all previous $\vec{q}_i$:

$$\vec{s}_k = \vec{a}_k - \sum_{i=1}^{k-1} \langle \vec{a}_k, \vec{q}_i \rangle \vec{q}_i, \qquad \vec{q}_k = \frac{\vec{s}_k}{\|\vec{s}_k\|_2}$$

Geometric picture: orthogonalizing two vectors in $\mathbb{R}^2$.



5.2 QR Decomposition

Gram–Schmidt gives us a matrix factorization. The key insight is that each original vector $\vec{a}_k$ can be written as a linear combination of the orthonormal vectors $\vec{q}_1, \ldots, \vec{q}_k$.

Recovering $\vec{a}_k$ from $\vec{q}_k$. By rearranging the Gram–Schmidt equations:

Column 1: Since $\vec{q}_1 = \vec{a}_1 / \|\vec{a}_1\|_2$, we have

$$\vec{a}_1 = \|\vec{a}_1\|_2 \, \vec{q}_1$$

Column 2: Since $\vec{s}_2 = \vec{a}_2 - \langle \vec{a}_2, \vec{q}_1 \rangle \vec{q}_1$ and $\vec{q}_2 = \vec{s}_2 / \|\vec{s}_2\|_2$, we get $\vec{s}_2 = \|\vec{s}_2\|_2 \, \vec{q}_2$, so

$$\vec{a}_2 = \langle \vec{a}_2, \vec{q}_1 \rangle \vec{q}_1 + \|\vec{s}_2\|_2 \, \vec{q}_2$$

Column 3: Similarly,

$$\vec{a}_3 = \langle \vec{a}_3, \vec{q}_1 \rangle \vec{q}_1 + \langle \vec{a}_3, \vec{q}_2 \rangle \vec{q}_2 + \|\vec{s}_3\|_2 \, \vec{q}_3$$

Reading off the matrix R. Write $A = [\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_m]$ and $Q = [\vec{q}_1, \vec{q}_2, \ldots, \vec{q}_m]$. The column relations above become $A = QR$, where:

$$A = QR, \qquad R = \begin{pmatrix} r_{11} & r_{12} & r_{13} & \cdots \\ 0 & r_{22} & r_{23} & \cdots \\ 0 & 0 & r_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Schematic view of $A = QR$:

$$\underset{A}{\boxed{\begin{pmatrix} \vec{a}_1 & \vec{a}_2 & \vec{a}_3 \end{pmatrix}}} = \underset{Q}{\boxed{\begin{pmatrix} \vec{q}_1 & \vec{q}_2 & \vec{q}_3 \end{pmatrix}}} \underset{R}{\boxed{\begin{pmatrix} r_{11} & r_{12} & r_{13} \\ 0 & r_{22} & r_{23} \\ 0 & 0 & r_{33} \end{pmatrix}}}$$

$$\vec{a}_1 = r_{11}\vec{q}_1, \quad \vec{a}_2 = r_{12}\vec{q}_1 + r_{22}\vec{q}_2, \quad \vec{a}_3 = r_{13}\vec{q}_1 + r_{23}\vec{q}_2 + r_{33}\vec{q}_3$$

$$Q^\top Q = I$$

The entries of the upper-triangular matrix R come directly from the Gram–Schmidt process:

- **Diagonal entries:** $r_{kk} = \|\vec{s}_k\|_2$ (normalization factors)
- **Off-diagonal entries:** $r_{jk} = \vec{q}_j^\top \vec{a}_k$ for $j < k$ (projection coefficients)

The matrix Q has **orthonormal columns**, meaning $Q^\top Q = I$. This follows because $\vec{q}_i^\top \vec{q}_j = 0$ for $i \neq j$ (orthogonality) and $\vec{q}_i^\top \vec{q}_i = 1$ (normalization).

Why QR matters: The QR decomposition converts the normal equations $A^\top A\vec{x} = A^\top \vec{y}$ into the simpler system $R\vec{x} = Q^\top \vec{y}$, which is easy to solve by back-substitution. This is numerically more stable than computing $A^\top A$ directly.

Derivation: Since $A = QR$ and $Q^\top Q = I$, we have $A^\top A = R^\top R$ and $A^\top \vec{y} = R^\top Q^\top \vec{y}$. The normal equations become $R^\top R\vec{x} = R^\top Q^\top \vec{y}$. Since R is invertible (each $r_{kk} > 0$), cancel $R^\top$ to get $R\vec{x} = Q^\top \vec{y}$.

5.3 Big Picture

This lecture developed two themes that reappear throughout the course.

Norms and dual norms. Optimizing a linear function over a norm ball reveals the dual norm:

$$\max_{\|\vec{x}\|_p \le 1} \vec{x}^\top \vec{y} = \|\vec{y}\|_q, \qquad \frac{1}{p} + \frac{1}{q} = 1$$

The three key dual pairs are:

- $\ell_1 \leftrightarrow \ell_\infty$: the dual of the ℓ_1 norm is the ℓ_∞ norm
- $\ell_2 \leftrightarrow \ell_2$: the ℓ_2 norm is self-dual
- $\ell_\infty \leftrightarrow \ell_1$: the dual of the ℓ_∞ norm is the ℓ_1 norm

Orthogonality and QR. The Gram–Schmidt process converts any set of linearly independent vectors into an orthonormal set spanning the same subspace. Recording the coefficients in a matrix yields the QR decomposition $A = QR$, which provides a numerically stable route to solving least-squares problems.

Key Takeaways

1. **A norm measures vector size:** it must satisfy non-negativity, the triangle inequality, and absolute homogeneity.
2. **The ℓ_p norms** form a family parameterized by p: $p = 1$ (sum of absolute values), $p = 2$ (Euclidean), $p = \infty$ (max component).
3. **Cauchy–Schwarz** bounds the inner product: $|\vec{x}^\top \vec{y}| \le \|\vec{x}\|_2 \|\vec{y}\|_2$. Hölder generalizes this to all ℓ_p norms.
4. **Minkowski's inequality** ($\|\vec{u} + \vec{v}\|_p \le \|\vec{u}\|_p + \|\vec{v}\|_p$) proves the triangle inequality for ℓ_p norms via Hölder.
5. **Norm balls** visualize constraints: diamond ($p = 1$), circle ($p = 2$), square ($p = \infty$).
6. **Dual norm pattern:** $\max_{\|\vec{x}\|_p \le 1} \vec{x}^\top \vec{y} = \|\vec{y}\|_q$ where $\frac{1}{p} + \frac{1}{q} = 1$. The general maximizer is $x_i^\star = \operatorname{sign}(y_i)|y_i|^{q-1}/\|\vec{y}\|_q^{q-1}$.
7. **Two proof strategies:** *Separability* (decouple into independent scalar problems) and *Bound-and-Attain* (establish an upper bound via an inequality, then construct a feasible point achieving it).
8. **Gram–Schmidt** converts linearly independent vectors to an orthonormal basis, yielding the QR decomposition $A = QR$.